

Alexandre Plana

Front-end / full-stack developer — Vue 3 · React · TypeScript

Industrial applications · Micro-frontends · AI

About

Thirteen years of industrial applications, from real-time 3D to the web. At Schneider Electric, I built several Vue 3 micro-frontends of the IECA IIoT platform, including the shared template, and held a cross-team role on shared components and code review. On Xistar, I cover design, front-end, part of the Go back end and delivery, working with AI agents I design and orchestrate.

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Languages

French Native · English Technical

Core Skills

FRONT-END

vue 3 · pinia · react · typescript · single-spa · tailwind

AI

claude · codex · ai agents

BACK-END

go · mqtt · rest api

QUALITY & DELIVERY

vitest · sonarqube · docker · gitlab ci

DESIGN & 3D

figma · unity 3d · c#

Interests

🎮 gaming · 🚵 mountain biking · 🥾 hiking · 🍳 cooking

Online resume
alexandre-plana.github.io/resume/

📌 Career overview

Datacorp — Mar 2026 – present — R&D Consultant — client X-PERT

x-pert/xistar — Full-stack developer

Xistar — maritime tactical command center · 2026

👥 3 developers + AI agents · from design to delivery

web · multi-screen · real-time · react · typescript · go · docker · gitlab ci

x-pert/ai-workflow — AI agent design & orchestration

AI-agent development workflow · 2026

ai · claude · codex · bmad · docker · gitlab ci

x-pert/xpert-design-system — UI design & front-end architecture

X-PERT design system · 2026

web · ux/ui · figma · react · shadcn · npm

Datacorp — 2017 – 2026 — R&D Consultant — client Schneider Electric

schneider-electric/iiot-edge-ieca — Front-end developer

Industrial Automation Edge (IECA) · 2023 – 2026

👥 4 Scrum teams · ~80 people · cross-team role

web · vue 3 · typescript · single-spa · pinia · vitest · sonarqube

schneider-electric/aoa-ar-app — Front-end, then lead dev

Augmented Operator Advisor (AOA) — Industrial AR Application · 2017 – 2023

👥 3 Scrum teams · front-end, then lead dev

3d / xr · unity · c# · ar · mobile · offline

Oreka Ingénierie — 2013 – 2017 — C# / Unity 3D Developer

suez/radioprotection — C# developer / Training simulation

Radiation-safety training · 2014 – 2015

serious game · radiation safety · nuclear · unity · c#

simulhom — Functional design & UX/UI

Musculoskeletal disorder prevention · 2015 – 2017

ux/ui · ergonomics · 3d simulation · unity · c#

chu-caen/emsim-basic — C# developer / 3D medical simulation

Radiation safety in interventional surgery · 2013 – 2015

3d simulation · radiation safety · medical · unity · c#

areva/emem — C# developer / 3D simulation

EMEM — mobile equipment evacuation enclosure · 2014 – 2016

3d simulation · nuclear · unity · c# · wpf

suez/maintenance-generateurs-de-vapeurs — C# developer / Serious game

Steam-generator maintenance · 2014 – 2015

serious game · nuclear · unity · c#

… and more serious games and simulators

📌 Vue.js · micro-frontend — Schneider Electric

Datacorp `2023 – 2026`

R&D Consultant — client Schneider Electric

📁 schneider-electric/iiot-edge-ieca

Front-end developer

Industrial Automation Edge (IECA) · 2023 – 2026

👥 4 Scrum teams of about ten people, around 80 people including validation and test automation

IECA is Schneider Electric's IIoT Edge platform: some fifteen Vue 3 micro-frontends loaded by single-spa, backed by Go services. I built five or six of them plus the shared template, and took part in developing and evolving each one listed below.

KEY CONTRIBUTIONS

- Built five or six micro-frontends and the shared template, then evolved them all in `#vue3`, `#typescript` and `#pinia`: connectivity, computations, conditions, alerts, scripting, data and administration.
- Cross-team role: shared components in the common library (configurable table, property selector, routing foundation) and code reviews, including pull requests from other teams.
- Quality focus: `#vitest` and `#jest` unit tests for components, stores and utilities, `#sonarqube` analysis; a clean codebase with very few reported defects.
- Occasional back-end extensions needed by front-end changes.

`vue3` `typescript` `single-spa` `pinia` `vitest` `jest` `sonarqube` `docker` `figma` `jira`

IECA MICRO-FRONTENDS I WORKED ON

All in `#vue3` and `#typescript`, loaded as `#single-spa` applications, with `#pinia`, Vue Router and Schneider Electric's Quartz design system. The newer applications share one architecture: container/presentational components, typed API-to-model mappers, one store per domain and forms validated with vee-validate and yup or zod.

SHARED FOUNDATION	
🧩 `@ieca/ieca-web-utils` **Shared library**	Configurable table, property selector, step wizard, breadcrumb header, dialogs; common stores, HTTP client, typings and test utilities.
🧩 `ieca-vue3-ui-template` **Micro-frontend template**	Starting point for new applications: rights-aware router, common stores, single-spa setup and tests.
DATA & CONNECTIVITY	
🧩 `ieca-connectivity-ui` **OT/IT connectivity**	The largest application: OPC UA and Modbus connectors configured through generated forms, a variable-editing grid, CSV import, IT interfaces and groups.
🧩 `ieca-data-management-ui` **Data management**	Filterable data-model tree, properties and their assignments, import, exportable history and value writes.
🧩 `ieca-datamodel-ui` **Data model**	Tree editor for the model: levels, property mapping and import; a rewrite of the former application for the new API.
PROCESSING & MONITORING	
🧩 `ieca-computation-ui` **Computations**	Computed variables created step by step: properties, server-validated formula with a function catalog, sliding or tumbling time window.
🧩 `ieca-condition-ui` **Conditions**	If/Then rules step by step: properties and quality level, validated formula, then an action (application, command or script).
🧩 `ieca-scripting-ui` **Scripting**	Scripts edited in Monaco with suggestions, run or cancelled; executions and parsed logs; rights-aware navigation.
🧩 `ieca-alert-manager-ui` **Alerts**	Alert list and history, acknowledgement and closing, bulk deletion and CSV export.

PLATFORM ADMINISTRATION	
🧩 `ieca-platform-configuration-ui` **Configuration**	Server certificate, disk usage, configuration export and import, application settings through generated forms.
🧩 `ieca-mail-ui` **SMTP mail**	SMTP server configuration with certificate checks and a test message.
🧩 `ieca-overview-ui` **Overview**	Installed Edge applications with their provider and license.
PROTOTYPES	
🧩 `schneider-electric/ieca-ip-sprint` **IP Sprints · 2024**	IECA data management on mobile (Capacitor then Quasar, Android and iOS); IECA alerts relayed to an on-call app as push notifications.

📌 Web development – X-PERT

Datacorp `Mar 2026 - present`
R&D Consultant – client X-PERT

📁 x-pert/xistar

Full-stack developer

Xistar – maritime tactical command center · 2026

👥 Team of 3 developers working with AI agents, multidisciplinary roles

A browser-based command center: tactical situation, mission and drone control, alerts, multi-screen operation and replay. I work from functional design to delivery, mostly on the front end.

KEY CONTRIBUTIONS

- Functional design, mockups, visual design and architecture decision records (ADRs) from operational requirements.
- Operator interface in `#typescript` and `#react`: `#dockview` workspace, `#maplibre` and `#deckgl` mapping with NATO APP-6 symbology.
- Multi-screen application driven from a master station: screen sets, remote view dispatch and real-time control of each display; time-based replay for debriefings.
- `#go` services, to a lesser extent: AIS, NMEA, CoT and MAVLink ingestion, publication over `#mqtt` and REST, alerts with escalation.
- End-to-end tests, `#docker` images and `#gitlab-ci` pipelines through to delivery.

Companion tools: a mobile app for alerts and commands (`#react`, `#socketio`), a video-wall emulator, and a drone and fleet simulator (`#nodejs`, `#mqtt`).

`typescript` `react` `go` `dockview` `maplibre` `deck.gl` `mqtt` `docker` `gitlab ci`

📁 x-pert/ai-workflow

AI agent design & orchestration

AI-agent development workflow · 2026

An AI-assisted development workflow inspired by the BMAD method and used on Xistar. Claude and Codex work in tandem: one produces, the other reviews.

KEY CONTRIBUTIONS

- Domain-specific agents and an orchestrator that dispatches work to them.
- Non-regression agents built on user journeys and a strictly defined design system.
- Isolated execution in one `#docker` container per project, with runs dispatched by `#gitlab-ci` to a dedicated machine.
- A `#react` and `#typescript` tracking tool fed by agent hooks.
- Session reports analyzed to correct agent instructions and the workflow.

`claude` `codex` `bmad` `docker` `gitlab ci` `react` `typescript`

📁 x-pert/xpert-design-system

UI design & front-end architecture

X-PERT design system · 2026

A design system shared by X-PERT's technical applications, website, documents and videos.

KEY CONTRIBUTIONS

- Designed components and interfaces in `#figma`.
- Built a `#react` and `#typescript` library on `#shadcn`: common components and technical visualizations (charts, treemaps, timelines, mapping, resizable panels).
- Distributed it through a private `#npm` registry, with integration documentation.

figma · react · typescript · shadcn · npm

📌 Industrial AR application – Schneider Electric

Datacorp

2017 – 2023

R&D Consultant – client Schneider Electric

📁 schneider-electric/aoa-ar-app

Front-end, then lead dev

Augmented Operator Advisor (AOA) – Industrial AR Application · 2017 – 2023

👥 3 Scrum teams (7, 5 and 4 people); joined as front-end developer, left as lead developer

Augmented-reality application for industrial operators, in C# with Unity, on Android, iOS and Windows. I was responsible for design, mockups, front-end architecture, development, releases and client-specific development.

KEY CONTRIBUTIONS

- Designed the interface, mockups and front-end architecture for three platforms, in portrait and landscape.
- Developed offline operation and `#http` synchronization, and contributed to the `#python` project server.
- Kept projects backward compatible by reading legacy formats and migrating them on load.
- Built an externalized style system and client-specific features for `#oem` customers.
- Shipped releases to each store and built a unit-testing tool wired into `#unity` build steps.

unity · c# · ar · android · ios · windows · python · figma · jira

📌 Serious games & simulation – Oreka Ingénierie

Oreka Ingénierie

2013 – 2017

C# / Unity 3D Developer

📁 oreka/xr-projects

Design & front-end development

Serious games and simulators – EDF, RTE, AREVA, DCNS, SUEZ · 2013 – 2017

👥 Teams of 1 to 3 people

Serious games, simulators and visualization tools for industry, nuclear and defense. My work focused on the design and front-end development of the applications.

KEY CONTRIBUTIONS

- Analyzed requirements and operating procedures and turned them into workflows, scenarios and mockups.
- Built 2D/3D interfaces and interactions in `#csharp` with `#unity`, including `#vr` and `#ar` devices.
- Integrated and optimized CAD models (`#solidworks`, `#3dsmax`) and CPU/GPU performance, through to delivery and client demos.

unity · c# · wpf · vr · uiux

PROJECTS

🗂️ **oreka/dosicase**
Dosicase · 2015 – 2017

Radiation-safety training simulator for practice sites: learning path, interface and interactions.

🗂️ `suez/radioprotection` **Radiation-safety training · 2014 – 2015**	Simulated accident scenarios and rescue-team management.
🗂️ `simulhom` **MSD prevention · 2015 – 2017**	3D ergonomics simulator designed with ergonomists: workflows, interface and postural-risk indicators.
🗂️ `chu-caen/emsim-basic` **Interventional surgery · 2013 – 2015**	Medical simulation: dosimetry and exposed areas shown throughout the scenarios.
🗂️ `areva/emem` **EMEM · 2014 – 2016**	3D simulation of a mobile enclosure for evacuating nuclear equipment, with WPF operator interfaces.
🗂️ `edf/aeroball` **Aeroball · 2014 – 2015**	Serious game for training on the Aeroball control station.
🗂️ `suez/maintenance-generateurs-de-vapeurs` **Steam generators · 2014 – 2015**	Serious game preparing maintenance work during plant outages.

🛠️ Personal projects

🛠️ AI-ASSISTED DEVELOPMENT WORKFLOW `in service`

iasit

Iasit is a coordination tool for software development assisted by AI agents. I designed it to structure preparation, execution, verification, correction and acceptance, track bugs as findings, and turn recurring problems into learned rules. Relevant rules are fed back into future prompts to improve subsequent executions. The application combines `#react` and `#typescript` with a `#nodejs` runtime.

🗓 2026 · Product design & full-stack development

`react` `typescript` `nodejs` `zod` `sigma.js` `api rest` `mcp`

🛠️ 3D EDUCATIONAL SANDBOX `active`

vorrery

Vorrery is a 3D educational sandbox designed for learning through experimentation, observation and hands-on manipulation. Users build their own systems, adjust parameters and immediately see the consequences of their choices. The product combines free exploration, guided experiments and child-friendly journeys in a `#react` `#typescript` interface rendered with `#threejs` and `#react-three-fiber`.

🗓 2026 · Product design & 3D front-end development

`react` `typescript` `three.js` `r3f` `web workers`

🎓 Education

📚 **React · TypeScript** Datacorp · 2026

📚 **Polymorph** Final-year internship · 2013

🎓 **BTS in Visual Communication – Multimedia option** 3DI Laval · 2010–2012

📚 **TypeScript · Docker · Vue.js** Datacorp · 2023

🎓 **Bachelor's degree in real-time 3D** 3DI Laval · 2012–2013